Department of Information Technology

Course Title & Course Code:
Software Engineering (5IT03)

Class: Third Year
Semester: V

Name of the Course Teacher:
Prof. Ms. N. N. Ghuikar

Title of the Innovative Practice:
Hands-on Experience in HR Management System using OrangeHRM (Open Source)

Objectives of the Practice:

- To provide students with practical exposure to software engineering concepts through a real-world HRM application.
- To enable students to explore open-source software tools for better understanding of requirement analysis, design, development, and testing.
- To enhance students' knowledge in modular design, database connectivity, and user interface interactions.
- To create an experimental feel for students by allowing them to perform HR operations (Employee management, Leave management, Attendance) within OrangeHRM.
- To bridge the gap between theoretical concepts of Software Engineering and their application in enterprise solutions.

Overview of the Practice:

Platform Used: OrangeHRM (Open-Source Version)
Modules Explored:

- Employee Information System
- Leave Management System
- Attendance Tracking

- Recruitment & Performance (selected modules)

Coverage:

- Software Engineering life cycle phases (Requirement, Design, Implementation, Testing, Maintenance).
- Case study of OrangeHRM functionalities mapped with SE concepts.

Steps of Implementation:

Step 1: Introduction & Installation

- Students were introduced to OrangeHRM open-source software.
- Installation on local systems / server environment was demonstrated.

Step 2: Requirement & Use Case Analysis

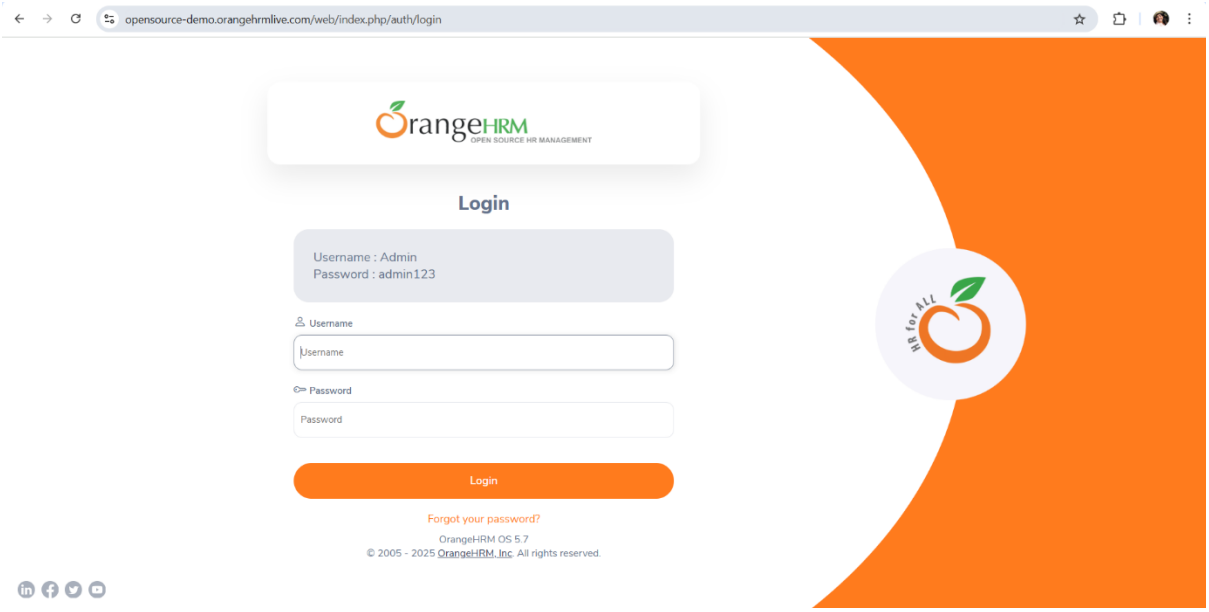
- HRM requirements were discussed in terms of use cases and system specification.
- Students prepared use case diagrams and requirement documents.

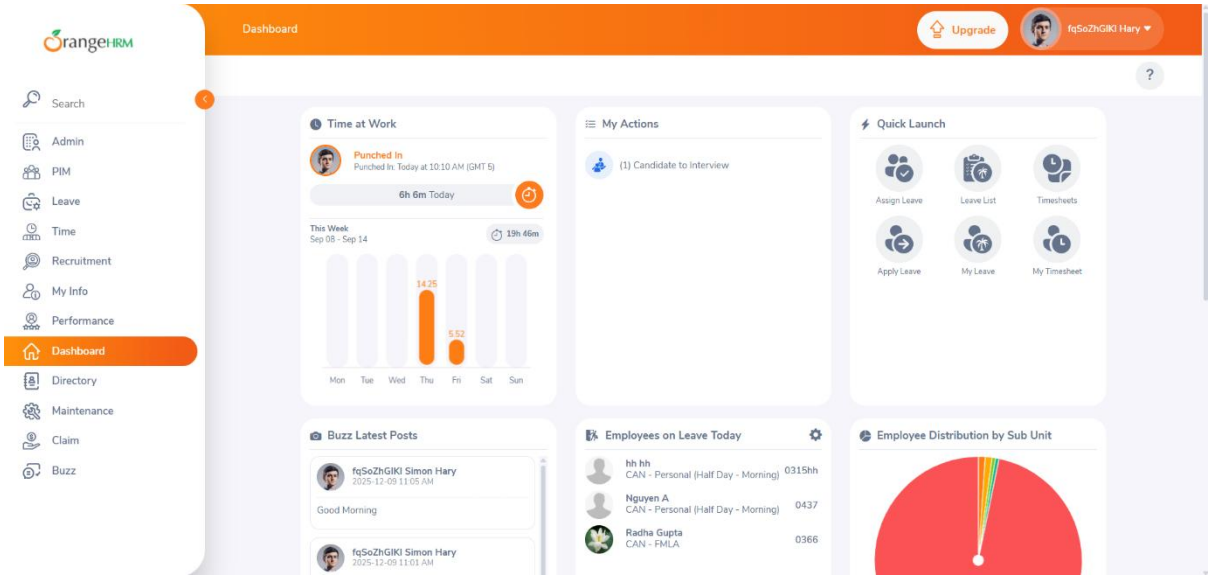
Step 3: Hands-On Exploration

- Students performed basic HR tasks like adding employees, managing leaves, checking attendance, and generating reports.
- Mapping of tasks with Software Engineering concepts (e.g., database design, modularity, testing).

Step 4: Testing & Evaluation

- Students tested OrangeHRM features (functional & non-functional).
- Identified limitations, bugs, and areas of improvement.





Outcomes and Impact:

- **Engagement:** Students experienced a real-time HRM software, increasing curiosity and interest.
- **Practical Understanding:** Students connected SE theoretical concepts (SRS, UML diagrams, testing strategies) with actual system operations.
- **Skill Development:** Improved ability in requirement analysis, system documentation, and evaluation of software quality.
- **Career Relevance:** Exposure to OrangeHRM boosted awareness of enterprise software used in IT and HR domains.

Advantages of the Initiative:

1. Provides industry-like exposure within an academic setting.
2. Students learn by exploring a live open-source project.
3. Enhances understanding of software lifecycle with practical examples.
4. Encourages team-based learning and project-oriented thinking.

POs Mapped:

PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PO12

Reflective Critique:

- Certain modules were complex; simplified tutorials improved accessibility.
- Group-based exploration helped students overcome technical difficulties collaboratively.

Challenges:

- Monitoring individual engagement in practical tasks.
- Limited lab session time to explore all modules in depth.